

Group Problem Set 18: Conservation of Energy in Circular Motion

1. A 10 kg ball is suspended above the ground on a pendulum where the pendulum arm is 15 m long. The ball is pulled back to a 45° angle θ and held in place, 4.4 meters above the plane. At a time $t=0$, the ball is released and starts to swing down and then back up to the other side of the pendulum. Below are 3 diagrams demonstrating the position of the pendulum at 3 different points in time. Use these diagrams to answer the following questions and assume no air resistance and negligible mass for the string.

time = 0

time = 1s

time = 2s

(a) What is the height of the pendulum at time = 1s? $\cos(45) = \frac{a}{15} \Rightarrow a = 10.6m \quad h_1 = 0.006$

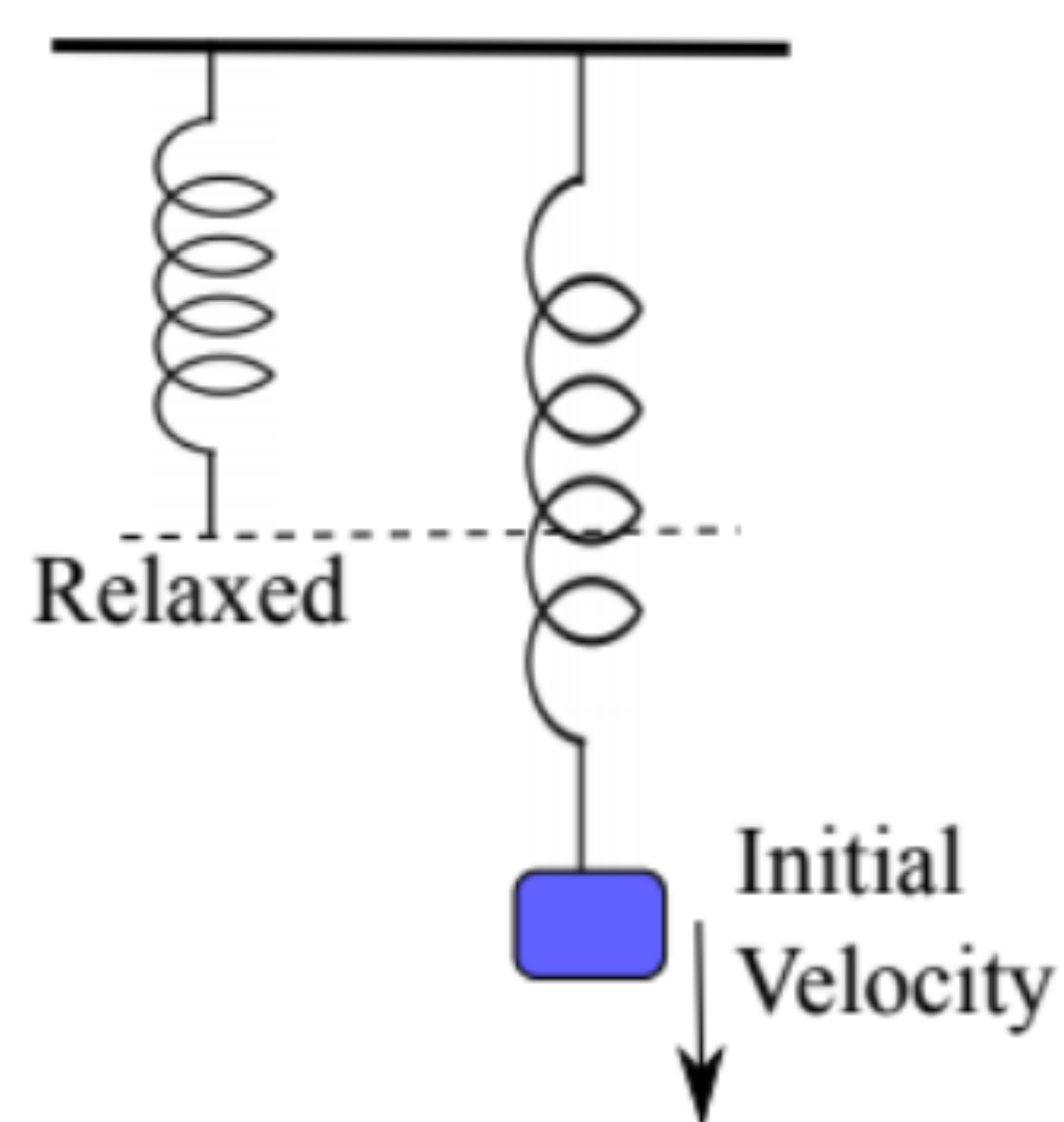
(b) What is the velocity of the ball at time $t=1s$? $v = \sqrt{2gh} \quad h = 4.4 - 0.006 \quad v = 9.37 \text{ m/s}$

(c) What is the height at 2s if the velocity there is 0? $h_2 = 4.4$

(d) Now let's add in air resistance. What is the work due to air resistance if the ball only reached 2/3 of its initial height at 2 s.

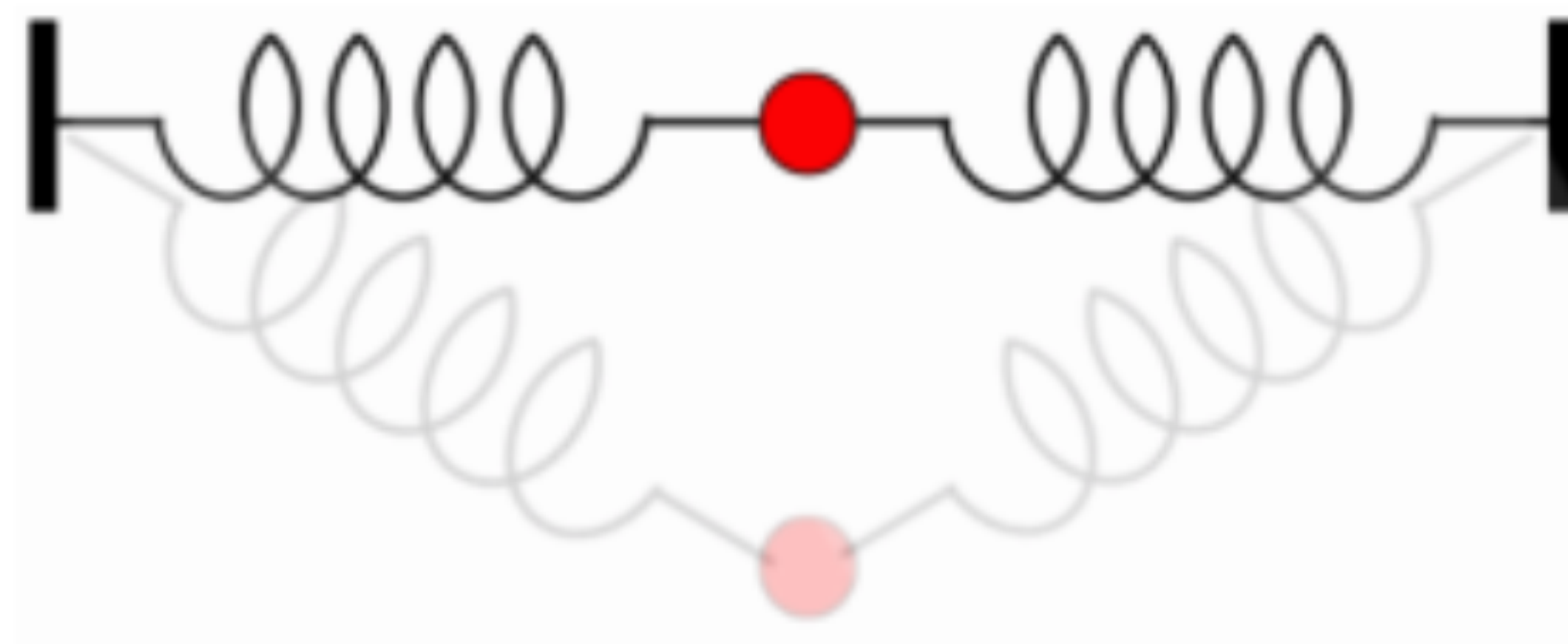
$E_f = E_i - W_{air}$

2. A block of mass 2 kg is attached to a spring of constant 100 N/m and the system is hung vertically. Initially the mass is pulled down, stretching the spring by 10 cm (from its natural resting place). You then further displace it by throwing it at 2 m/s in the downward pointing direction.



- (a) Find the speed of the block when the spring returns to its relaxed state.
- (b) Find the lowest and highest points that the block reaches.
- (c) Does the answer depend on the direction that you throw the block?

3. Two identical springs with constants 50.0 N/m are attached to a block of mass 1.0 kg . The other ends of the springs are attached to two parallel walls, as shown in the figure. Initially the springs are horizontal and unstretched, with the unstretched length of each spring equaling 25.0 cm . The block is released from rest.



- (a) Find the speed of the block when it has fallen vertically 10 cm .
- (b) Find the vertical position of the lowest point the block reaches.

4. An object of mass 10 kg moves in a straight line and experiences a force of $F_x = (-10\frac{N}{m})x^2$.
- (a) What is the work done by this force as the object moves from $x = 1m$ to $x = 5m$?
 - (b) At $x = 1m$ the object has a speed of $10m/s$. What is the object's mass?
 - (c) What is the speed of the object when it reaches $x = 5m$?
 - (d) How far from its initial position, is the object when it stops?
 - (e) Using excel, can you plot the force v. distance?
 - (f) How much work is done within the first 1 meter?
 - (g) How much work is done on the object in the last 1 meter?